Surfaces and their Areas

Lecture 09

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Surface Integrals

Inquiry Questions:

- To compute the mass of a surface
- To compute the flow of a liquid across a curved membrane
- To compute the total electrical charge on a surface

- We need to integrate a function over a curved surface in space.
- Such a **surface integral** is the two-dimensional extension of a the line integral concept used to integrate over a one-dimensional curve.
- Like line integrals, surface integrals arise in two forms:
  (a) One form occurs when we integrate a scalar function to find its total mass. This is known as **scalar surface integral**.
  (b) The second form is for surface integrals of vector fields, analogus to the line integrals for vector fields known as **vector surface integral**.
  (c) An example of this form occurs when we want to measure the net flow of a fluid across a surface submerged in the fluid (just as we previously defined the flux of $\mathbf{F}$ across C).

Formulas for a Surface Integral of a Scalar Functione

1. For a smooth surface S defined parametrically as

$$\mathbf{r}(u,v) = f(u,v)\mathbf{i} + g(u,v)\mathbf{j} + h(u,v)\mathbf{k}, \quad (u,v) \in R,$$

and a continuous function $G(x,y,z)$ defined on S, the surface integral of G over S is given by the double integral over R,

$$\boxed{\iint_S G(x,y,z)\,d\sigma = \iint_R G(f(u,v), g(u,v), h(u,v))\,|\mathbf{r}_u \times \mathbf{r}_v|\,dudv.}$$

2. For a surface S given **implicitly** by $F(x,y,z) = c$, where F is a continuously differentiable function, with S lying above its closed and bounded shadow region R in the coordinate plane beneath it, the surface integral of the continuous function G over S is given by the double integral over R,

$$\boxed{\iint_S G(x,y,z)\,d\sigma = \iint_R G(x,y,z)\frac{|\nabla F|}{|\nabla F \cdot \mathbf{p}|}\,dA,}$$

where $\mathbf{p}$ is a unit vector normal to R and $\nabla F \cdot \mathbf{p} \neq 0$.

3. For a surface S given **explicitly** as the graph of $z = f(x,y)$, where f is a continuously differentiable function over a region R in the xy-plane, the surface integral of the continuous function G over S is given by the double integral over R,

$$\boxed{\iint_S G(x,y,z)\,d\sigma = \iint_R G(x,y,f(x,y))\sqrt{f_x^2 + f_y^2 + 1}\,dx\,dy.}$$

Examples

1. Integrate $G(x,y,z) = x^2$ over the cone $z = \sqrt{x^2+y^2}$, $0 \leq z \leq 1$.

2. Integrate $G(x,y,z) = xyz$ over the surface of the cube cut from the first octant by the planes $x = 1, y = 1$, and $z = 1$.

3. Evaluate $\iint_S \sqrt{x(1+2z)}d\sigma$ on the portion of the cylinder $z = \frac{y^2}{2}$ over the triangular region $R : x \geq 0, y \geq 0, x + y \leq 1$ in the xy-plane.